SmartInterview: Personalized Technical Interview Preparation in Higher Education

Slide 1: Title Slide

Project Name: SmartInterview — Personalized Technical Interview Preparation in Higher Education

Project Team ID: Team G-1392

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Category: Edu-Tech

Slide 2: Introduction & Problem Statement

The Challenge in Modern Recruitment:

- Engineering students face a stark disconnect between university curricula and dynamic industry job descriptions.
- Traditional preparation methods (static question banks, mock peers) fail to measure cognitive depth or adapt to the candidate's skill level.
- Existing literature reflects critical gaps:

* **Base Paper 1 (Nagarajan et al., Feb 2026):** Built for enterprise recruiter filtering; lacks student learning roadmaps, code sandboxes, or curriculum grounding.

* **Base Paper 2 (Wahid et al., March 2026):** Voice is strictly one-way (silent AI, candidate reads questions); rigid domain dropdowns without custom JD matching.

- **The Gap:** Candidates lack a low-latency, two-way conversational coach grounded in verified technical knowledge with objective evaluation.

Slide 3: Proposed Solution — SmartInterview

An AI-Powered, Adaptive Mock Interview Platform for Higher Education:

- **3 Operational Modes:**

1. *Job-Specific Mode:* Set-theoretic resume-to-JD mapping (Group A Overlap vs. Group B Gaps).
2. *Topic-Based Mode:* Granular technical drilling across chosen CS domains.
3. *Semester Syllabus Mode:* Isolated vector collections from university syllabus PDFs.

- **RAG-Grounded:** Retrieves from 402 curated CS chunks (Hit@1: 96.97%, Hit@3: 100%, MRR: 0.9798) to kill hallucinations.
- **Deterministic Cognitive Scaffolding:** State-driven difficulty scaling via Bloom's Revised Taxonomy.
- **Multi-Signal Evaluation:** 5-factor mathematical rubric eliminating subjective LLM "vibe grading".

Slide 4: System Architecture

Decoupled 4-Tier Distributed Technology Stack:

- **Presentation Tier:** React 18 SPA (Vite, Tailwind CSS, Web Speech API audio visualizer).
- **Application Tier:** FastAPI (Python 3.10+, Uvicorn ASGI, Pydantic v2, HS256 JWT auth).
- **Core Intelligence Tier:**

* ChromaDB HNSW vector store (**`technical_kb`**, 402 chunks, 384-dim S-BERT embeddings).

* Groq Cloud LPU high-speed inference (**`llama-3.3-70b-versatile`** / **`gpt-oss-120b`**, 320ms TTFT).

* Deterministic Python Bloom Adaptive Engine.

- **Voice Subsystem & Persistence:**

* Microsoft Edge-TTS (**`en-US-ChristopherNeural`**, in-memory **`io.BytesIO`**, <= 1.2s latency).

* Whisper Large-v3 STT with domain-primed technical vocabulary prompt.

* MySQL 8.0 relational database for users, sessions, and transcripts.

Slide 5: Core Workflow & Context Ingestion

How SmartInterview personalizes the experience:

1. **Document Ingestion:** Candidate uploads Resume (PDF) and target Job Description (or Syllabus).
2. **Deterministic Extraction:** PyMuPDF isolates technical skills and experience entities.
3. **Set-Theoretic Skill Gap Analysis:**
   - **Group A (Role Overlap):** Skills present in both Resume and JD -> Tested for verification.
   - **Group B (Role Gaps):** Skills required by JD but missing in Resume -> Tested for adaptability.
4. **Targeted Staging:** The interview dynamically focuses on validating Group A and bridging Group B.

Slide 6: Hybrid Graph RAG Architecture

Eliminating LLM Hallucinations & Enabling Multi-Hop Reasoning:

- **The Problem:** Vanilla vector search misses exact technical keywords (**`2PL`**, **`ACID`**) and cannot traverse conceptual prerequisites.
- **Tri-Retrieval Engine:**

* **Dense Vector Search (ChromaDB):** Semantic similarity matching via S-BERT embeddings.

* **Lexical Keyword Search (BM25Okapi):** Deterministic exact token matching for symbols, algorithms, and acronyms.

* **Knowledge Graph Traversal (NetworkX/Neo4j):** Directed graph mapping 50 CS concepts, prerequisite edges, and Bloom cognitive depths across 8 core domains (**`data/graph_data.json`**).

- **Reciprocal Rank Fusion (RRF) & Cross-Encoder Reranker:**

* Merges rankings via $RRF(d) = \sum \frac{1}{k + r(d)}$ and cross-encoder scoring.

* **Hit@1 = 96.97%**, **Hit@3 = 100.00%**, **MRR = 0.9798** across 33 domain benchmarks.

* Embedded in-memory graph traversal $< 12\text{ms}$; total pipeline $< 650\text{ms}$.

Slide 7: Deterministic Adaptive Progression (Bloom's Taxonomy)

Strict Cognitive Difficulty Scaling:

- Arbitrary LLM difficulty jumps replaced with a deterministic Python state machine mapped to **Bloom’s Revised Taxonomy** (*Remember, Understand, Apply, Analyze, Evaluate, Create*).
- **The State Transition Logic:**
- **Score >= 80% (Mastery):** Advance to higher Bloom cognitive level (+1) or advance difficulty tier.
- **Score 50% – 79% (Adequate):** Maintain current Bloom level and difficulty tier.
- **Score < 50% (Remediation):** Regress Bloom level (-1) or lower difficulty to rebuild foundations.
- **Weakness-Biased Skill Selection:** Engine automatically prioritizes candidate's lowest-scoring skills.

Slide 8: Multi-Signal Answer Evaluation

Objective, Multi-Dimensional Scoring (Replacing "Vibe Grading"):

- Evaluates candidate answers using a strict 5-part weighted mathematical formula:

* **30% Technical Correctness:** Deep LLM logical reasoning and accuracy analysis.

* **20% Completeness:** Coverage of all question constraints and edge cases.

* **20% Relevance:** Focus, conciseness, and absence of filler rhetoric.

* **15% Semantic Cosine Similarity:** S-BERT (**`all-MiniLM-L6-v2`**) matching candidate answer against verified ground-truth chunk.

* **15% Concept Coverage:** Deterministic detection of mandatory technical keywords and concepts.

- **Why?** Blends semantic understanding with hard concept verification, ensuring reproducible, unbiased marks.

Slide 9: Remedial Mastery Mode & Voice Synthesis

Transforming Assessment into Active Upskilling:

- **Remedial Mastery Mode:**

* **Knowledge Graph Root-Cause Tracing:** Automatically diagnoses missing prerequisite concepts when scores dip.

* **Dynamic Micro-Lessons & Flashcards:** Instantly generates interactive revision cards, conceptual summaries, and common pitfalls.

* **Targeted Retry Drills:** Enables candidates to immediately re-attempt failed concepts with adaptive scaffolding.

- **Two-Way Voice Experience:**

* Neural Edge-TTS voice streaming with in-memory **`io.BytesIO`** buffers ($\le 1.2\text{s}$ latency).

* Groq Whisper transcription (**`whisper-large-v3`**) with CS domain vocabulary priming.

* Auto-generates downloadable ReportLab PDF performance roadmaps with 5-axis radar charts.

Slide 10: System Verification & 274-Test Matrix

Comprehensive Boundary & Integration Validation:

- **274 Total Unit & Integration Tests across 18 Phases — 100% Passing:**

TC-01 (Auth): *Duplicate registration rejected (HTTP 400)* — PASSED*

TC-02 (Parsing): *Corrupt/invalid file upload rejected (HTTP 400)* — PASSED*

TC-03 (Hybrid RAG): *Tri-retriever RRF scoring & canonical deduplication* — PASSED*

TC-04 (Graph Store): *Automatic NetworkX in-memory fallback when Neo4j is absent* — PASSED*

TC-05 (Bloom Scaling): *Dynamic tier advancement ($\ge 80\%$) and regression ($< 50\%$)* — PASSED*

TC-06 (Remediation): *Prerequisite root-cause diagnosis and retry question dispatch* — PASSED*

TC-07 (Rubric Determinism): *5-factor mathematical formula convergence* — PASSED*

TC-08 (Failure Safeguards): *Candidate starvation prevention and corrupt query handling* — PASSED*

Slide 11: Conclusion & Questions

Conclusion:

SmartInterview bridges the gap between theoretical higher education and dynamic industry hiring. By combining Hybrid Graph RAG grounding (Hit@1: 96.97%, Hit@3: 100%), deterministic Bloom's taxonomy cognitive scaffolding, Remedial Mastery drills, and multi-signal objective evaluation, we deliver an authentic, unbiased, low-latency interview preparation ecosystem.

Thank You! We are now open for questions from the panel.